Characterize necessity of the I-property for isolated calmness of KKT mapping

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Title

A counterexample to the necessity of the I-property for isolated calmness of the KKT mapping in Nash equilibrium problems

Abstract

We show that, for a general smooth Nash equilibrium problem with canonical perturbations, isolated calmness of the KKT solution mapping at a reference KKT point does not imply the I-property stated in the problem. The counterexample is an unconstrained two-player quadratic game. Its KKT mapping is an affine single-valued map, hence isolated calm, while the I-property fails on the whole space.

Problem statement

The question is whether the following implication is valid for a general canonically perturbed Nash equilibrium problem:

$$S_{\mathrm{KKT}} \text{ is isolated calm at } \bar{p} \text{ for } (\bar{x}, \bar{\lambda}) \quad \Longrightarrow \quad \text{the I-property holds at } (\bar{p}, \bar{x}, \bar{\lambda}) \text{ on } K(I_1, I_2).$$

Equivalently, does isolated calmness of the KKT solution mapping force the condition

$$\forall y = (y_1, \ldots, y_N) \in K(I_1, I_2), \qquad \Big(\forall k,\ \sum_{i=1}^{N} y_k^T \nabla^2_{x_k x_i} L_k(\bar{x}_k, \bar{x}_{-k}, \bar{\lambda}_k; \bar{w}_k) y_i = 0\Big) \Longrightarrow y = 0?$$

Main result

Theorem 1. *The implication above is false in general.*

More precisely, there exists a smooth two-player Nash equilibrium problem with canonical perturbations for which the KKT solution mapping S_{KKT} is isolated calm at a reference parameter for a reference KKT point, but the I-property fails at that point.

Proof

Consider the case $N = 2$, $n_1 = n_2 = 2$, and no constraints:

$$m_1 = m_2 = 0.$$

Thus the multiplier component belongs to $\mathbb{R}^0$ and is vacuous. Introduce dummy parameters $w_1, w_2 \in \mathbb{R}$, but let the objectives be independent of them. For $v_1, v_2 \in \mathbb{R}^2$, define

$$f_1(x_1, x_2; w_1) = \frac{1}{2}\|x_1\|^2 + x_1^T A x_2, \qquad f_2(x_2, x_1; w_2) = \frac{1}{2}\|x_2\|^2 + x_2^T A^T x_1,$$

where

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 0 \end{pmatrix}.$$

The perturbed player problems are therefore

$$\min_{x_1 \in \mathbb{R}^2} \frac{1}{2}\|x_1\|^2 + x_1^T A x_2 - \langle v_1, x_1 \rangle, \qquad \min_{x_2 \in \mathbb{R}^2} \frac{1}{2}\|x_2\|^2 + x_2^T A^T x_1 - \langle v_2, x_2 \rangle.$$

Set

$$\bar{p} = (\bar{v}_1, \bar{v}_2, \bar{w}_1, \bar{w}_2) = (0,0,0,0), \qquad \bar{x} = (0,0).$$

Since there are no constraints, the Lagrangians are just the objectives:

$$L_1 = f_1, \qquad L_2 = f_2.$$

For this unconstrained game, the KKT system is only the stationarity system:

$$\nabla_{x_1} f_1(x_1, x_2) - v_1 = 0, \qquad \nabla_{x_2} f_2(x_2, x_1) - v_2 = 0.$$

Because

$$\nabla_{x_1} f_1(x_1, x_2) = x_1 + A x_2, \qquad \nabla_{x_2} f_2(x_2, x_1) = x_2 + A^T x_1,$$

the KKT system becomes

$$x_1 + A x_2 = v_1, \qquad A^T x_1 + x_2 = v_2.$$

Writing $x = (x_1, x_2) \in \mathbb{R}^4$ and $v = (v_1, v_2) \in \mathbb{R}^4$, this is

$$Mx = v, \qquad M = \begin{pmatrix} I_2 & A \\ A^T & I_2 \end{pmatrix}.$$

We now show that M is invertible. Since

$$A^T A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix},$$

the Schur complement with respect to the upper-left block I_2 is

$$I_2 - A^T A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

whose determinant is $-1 \neq 0$. Hence

$$\det M = \det(I_2) \det(I_2 - A^T A) \neq 0,$$

so M is invertible.

Therefore, for every parameter $p = (v_1, v_2, w_1, w_2)$,

$$S_{\mathrm{KKT}}(p) = \{(M^{-1}v, 0_{\mathbb{R}^0})\}.$$

In particular, $S_{\mathrm{KKT}}(\bar{p}) = \{(\bar{x}, 0_{\mathbb{R}^0})\}$. Moreover,

$$\|(M^{-1}v, 0_{\mathbb{R}^0}) - (\bar{x}, 0_{\mathbb{R}^0})\| = \|M^{-1}v\| \leq \|M^{-1}\| \, \|v\| \leq \|M^{-1}\| \, \|p - \bar{p}\|.$$

Thus S_{KKT} is isolated calm at $\bar{p}$ for $(\bar{x}, 0_{\mathbb{R}^0})$.

We next check the I-property. Since $m_1 = m_2 = 0$, all index sets I_1^k and I_2^k are empty. Hence

$$K(I_1, I_2) = \mathbb{R}^4.$$

Also,

$$\nabla^2_{x_1x_1} L_1 = I_2, \qquad \nabla^2_{x_1x_2} L_1 = A, \qquad \nabla^2_{x_2x_1} L_2 = A^T, \qquad \nabla^2_{x_2x_2} L_2 = I_2.$$

Let

$$e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad y_1 = e, \qquad y_2 = e, \qquad y = (y_1, y_2) \neq 0.$$

Then

$$I_2 e + Ae = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so

$$y_1^T \left(\nabla^2_{x_1x_1} L_1 \, y_1 + \nabla^2_{x_1x_2} L_1 \, y_2 \right) = e^T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

Likewise,

$$A^T e + I_2 e = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and therefore

$$y_2^T \left(\nabla^2_{x_2x_1} L_2 \, y_1 + \nabla^2_{x_2x_2} L_2 \, y_2 \right) = e^T \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0.$$

Thus $y \in K(I_1, I_2)$, $y \neq 0$, and for each player $k = 1, 2$,

$$\sum_{i=1}^{2} y_k^T \nabla^2_{x_k x_i} L_k(\bar{x}_k, \bar{x}_{-k}, \bar{\lambda}_k; \bar{w}_k) y_i = 0.$$

So the premise of the I-property holds for a nonzero y, which means that the I-property fails.
Hence isolated calmness of S_{KKT} does not imply the I-property in general. $\square$

Remarks

- The counterexample is fully smooth and unconstrained; hence the failure is not caused by pathological constraints.
- Each player's objective is strongly convex in its own decision variable, since the Hessian with respect to x_k is I_2. Thus the stationarity condition for each player is sufficient for optimality, and $\bar{x} = (0, 0)$ is in fact the unique Nash equilibrium at $\bar{p}$.
- Consequently, any valid converse implication from isolated calmness to the I-property must require additional hypotheses beyond isolated calmness alone.